

Introduction to Cloud Computing

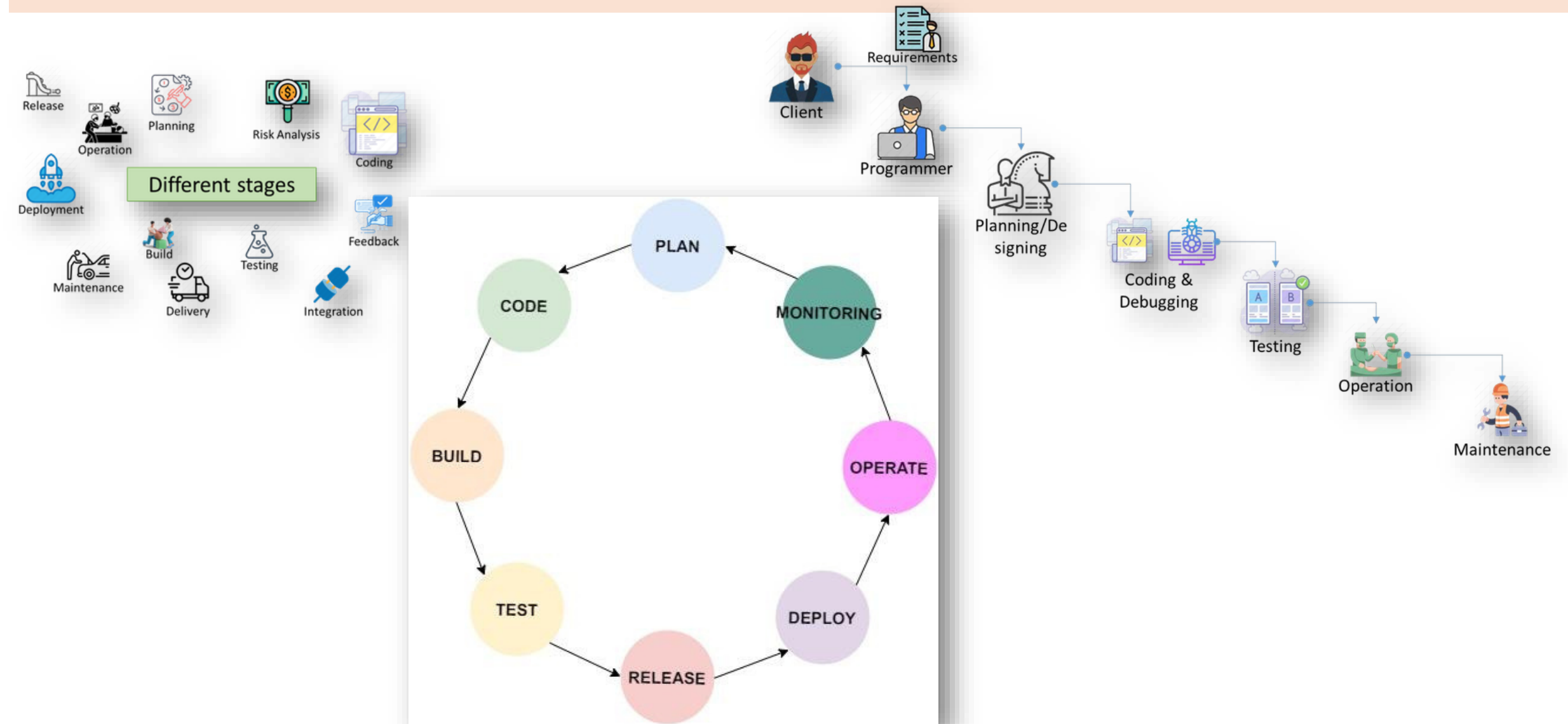
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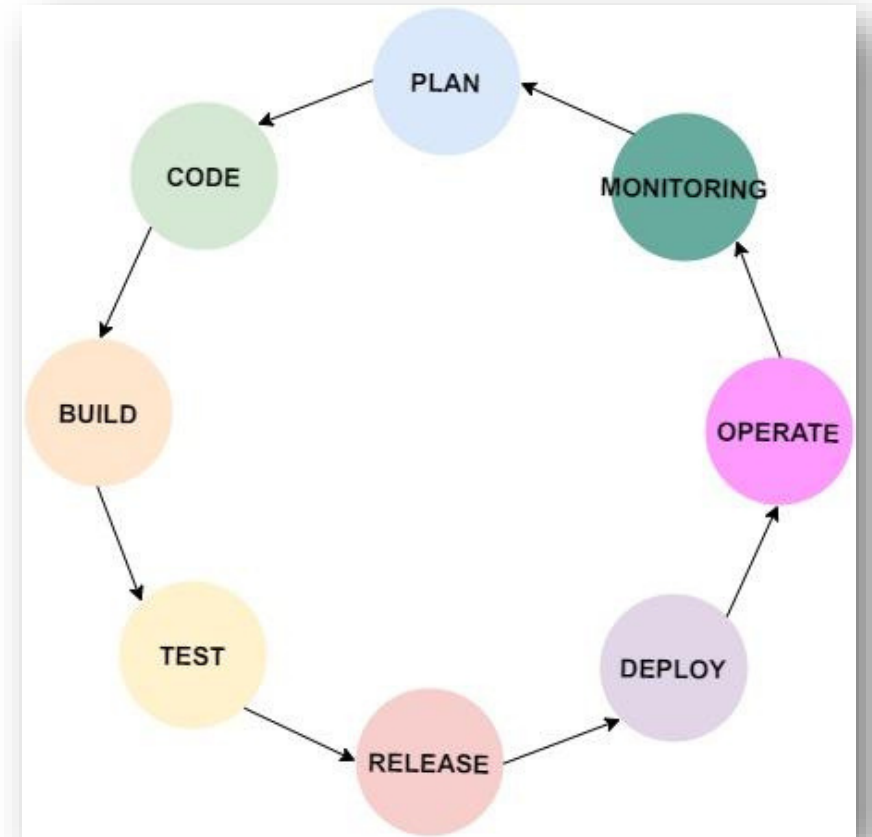
OUTLINE

- What is Cloud Computing?
 - Deployment models
 - Service models
 - Virtualization & Hypervisor

Recap



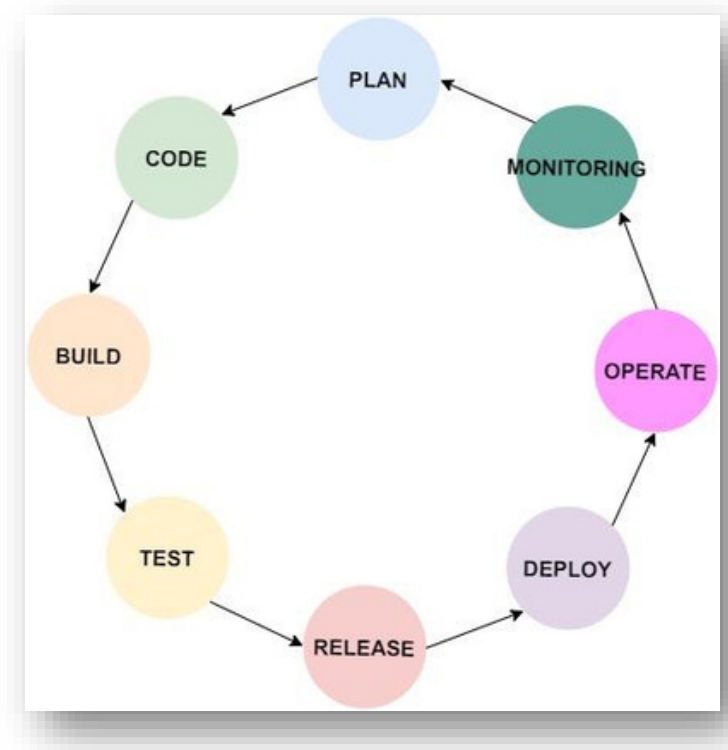
Cloud computing



Cloud computing ?

Why should I learn Cloud computing in this course?

- DevOps Tooling and Ecosystem
- Build-Test-deployment environment
- Resource Provisioning and Scalability
- Elastic environment
- Infrastructure automation
- CICD
- And many more....



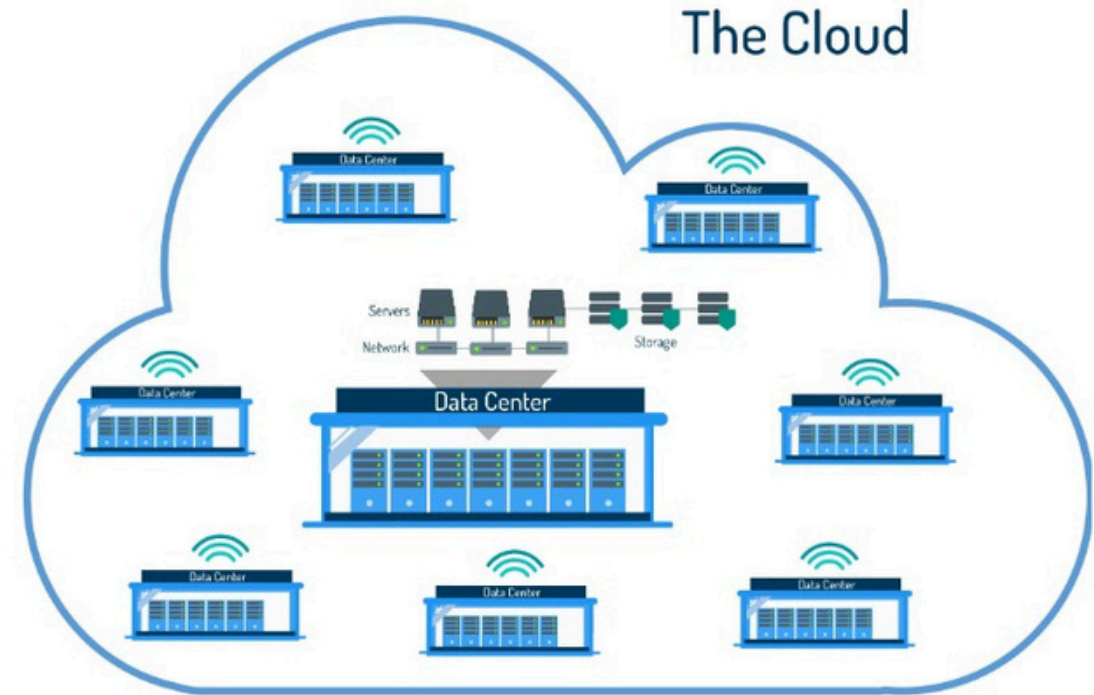
What is Cloud Computing?

- Computing as a utility
 - Utility services e.g. water, electricity, gas, etc.
 - Consumers pay based on their usage

According to NIST Cloud computing is **a model** for enabling *ubiquitous, convenient, on-demand network access* to a **shared** pool of **configurable** computing resources (e.g., networks, servers, storage, applications, and services) that can be **rapidly** provisioned and released with **minimal management** effort or service provider interaction. (Special Publication 800-145)

What is Cloud Computing?

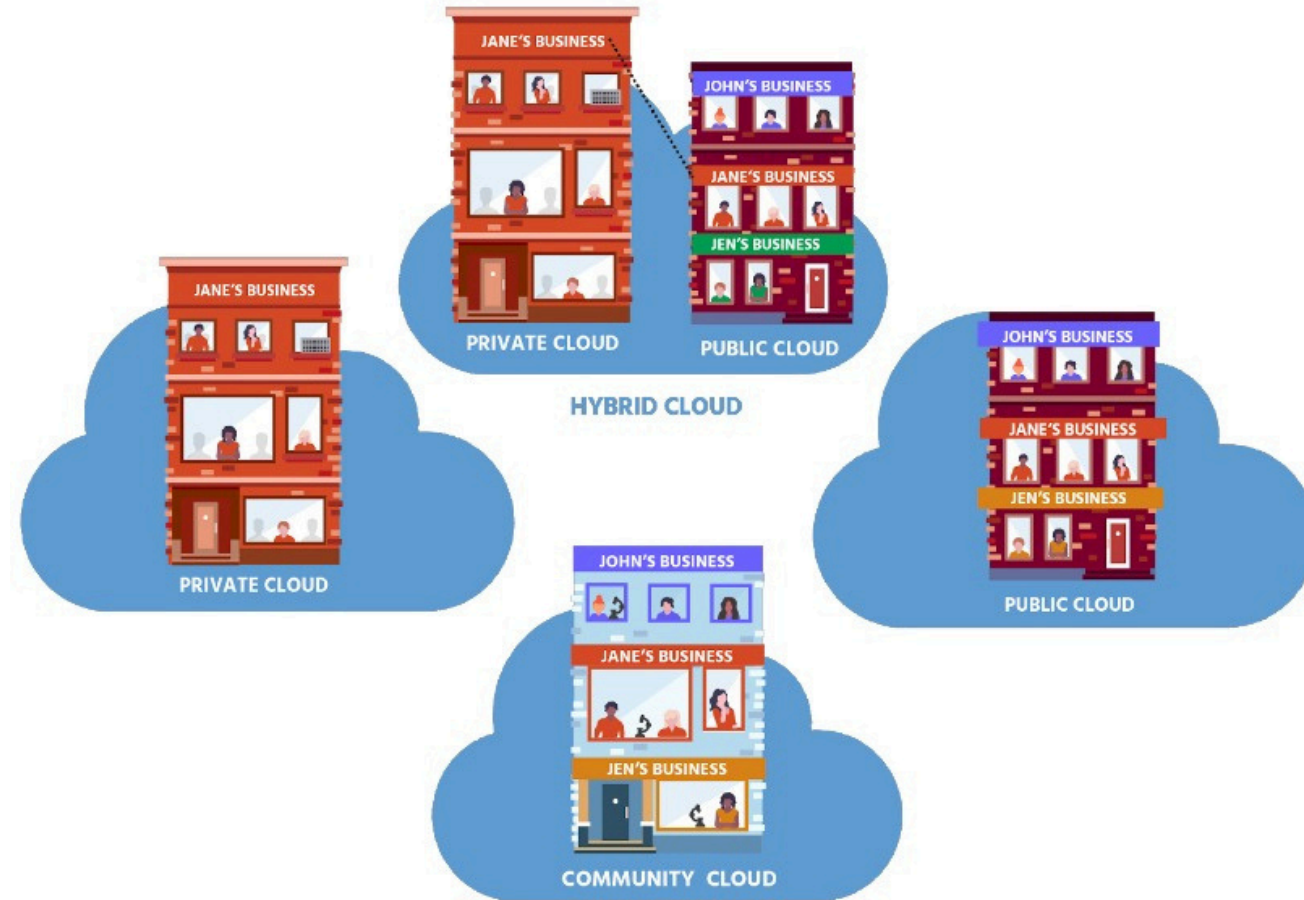
- Cloud computing is the delivery of shared computing resources (software and/or data) on-demand through the internet



Cloud deployment model

Cloud deployment includes an emphasize on **where** the hardware or software is running and **who** is controlling it.

- Private Cloud
- Community Cloud
- Public Cloud
- Hybrid Cloud



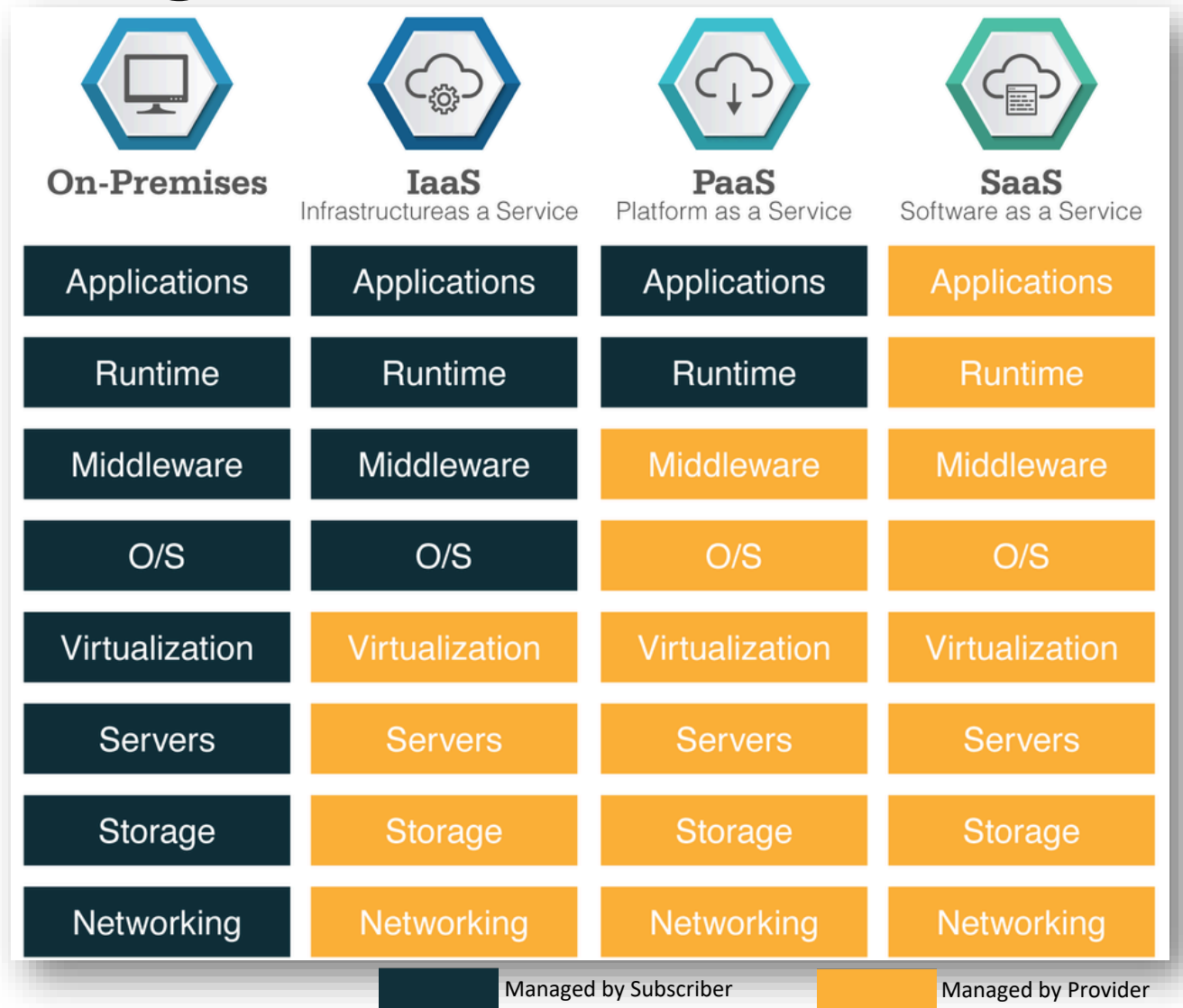
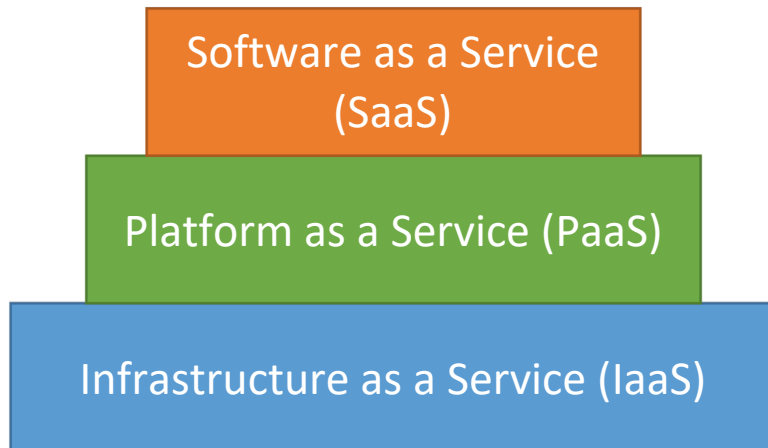
Cloud deployment models comparison

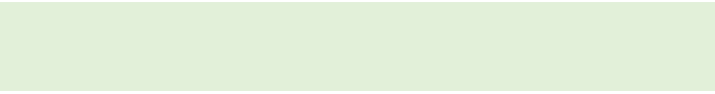
Description	Public Cloud	Private Cloud	HybridCloud
Cloud environment	Multi-Tenancy-Shared environment.	Single tenancy-only for single use of an organization.	Both single tenancy and multi –tenancy.
Data center location	Anywhere	Inside the organization's network.	Both
Resource sharing (Server hardware, network and storage)	shared by multiple users in the cloud.	No sharing of resources.	Very secure; integration options add an additional layer of security.

Types of Cloud Computing service models

Classification of the types of services and resources:

- Software as a Service (SaaS)
- Platform as a Service (PaaS)
- Infrastructure as a Service (IaaS)

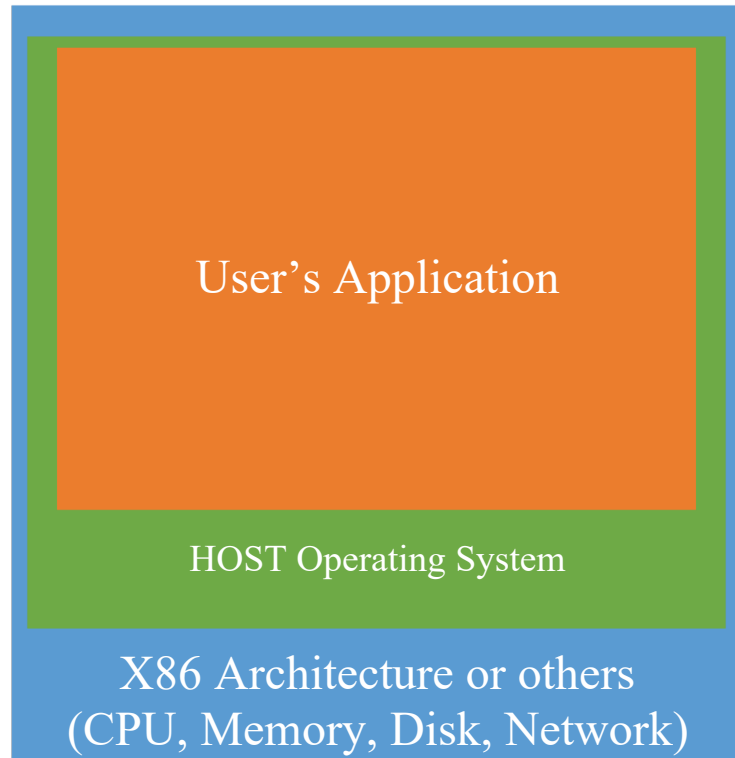




–The underlined technology

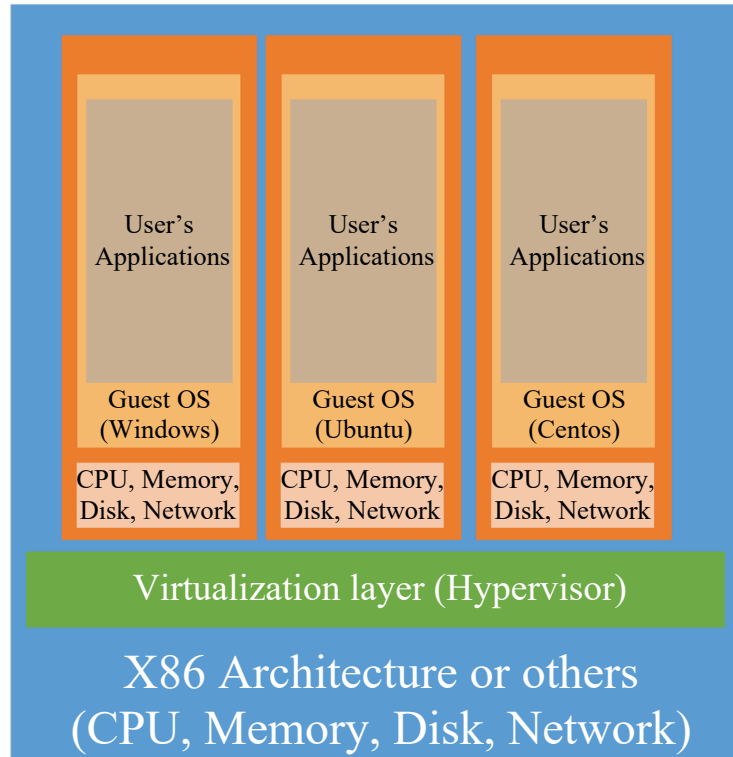
Virtualization –The underlined technology

Without Virtualization –traditional physical server



- Single OS per machine
- Single/multiple application(s) per machine
 - Database server
 - Web server
- Hardware components connected directly to operating system
 - **CPU**
 - **Memory**
 - **Disk**
 - **Network Card**

With Virtualization



Why Virtualization ?

- Increased **performance** and computing **capacity** relevance to hardware manufacturing.
- **Underutilized** Hardware and software Resources
- Lack of Space
- Greening Initiatives
- Rise of administrative costs
 - Administrative tasks include: labor intensive
 - hardware monitoring
 - defective hardware replacement
 - server setup and updates
 - resource monitoring backups

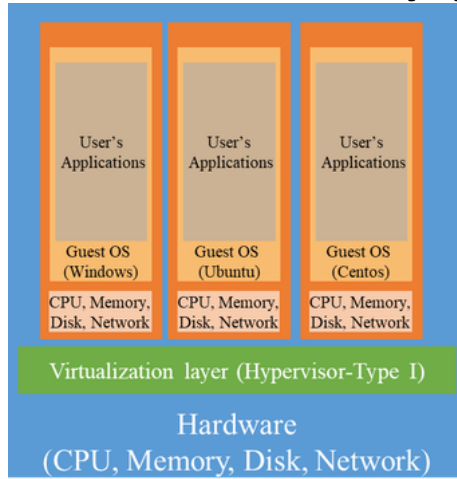
Virtualization

- To define it in a general sense,
 - virtualization encompasses any technology -either software or hardware -that adds an extra layer of isolation or extra flexibility to a standard system.
- Virtualization is mainly used to **emulate** *execution environment*, storage and networks.
- Execution Environment classified into two :
 - Process-level –implemented on top of an existing operating system.
 - System-level –implemented directly on hardware with no or minimum requirement of existing operating system
- Familiar examples include
 - virtual memory for memory management,
 - virtual disks to allow for partitioning

Hypervisor

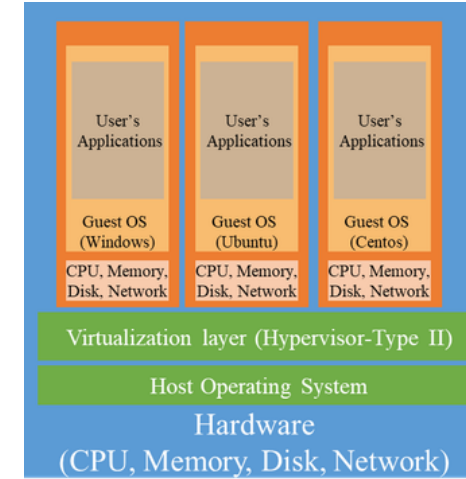
- *a software program used to run and manage one or more virtual machines on a computer.* [[src](#)]
- It recreates a h/w environment.

Hypervisor type



Type-I

- It runs directly on top of the hardware.
- Also called *bare metal*
- Takes place of OS.
- Directly interact with the ISA exposed by the underlying hardware.
- Also known as *native virtual machine*.
Example: VmWare ESXi, MS HyperV



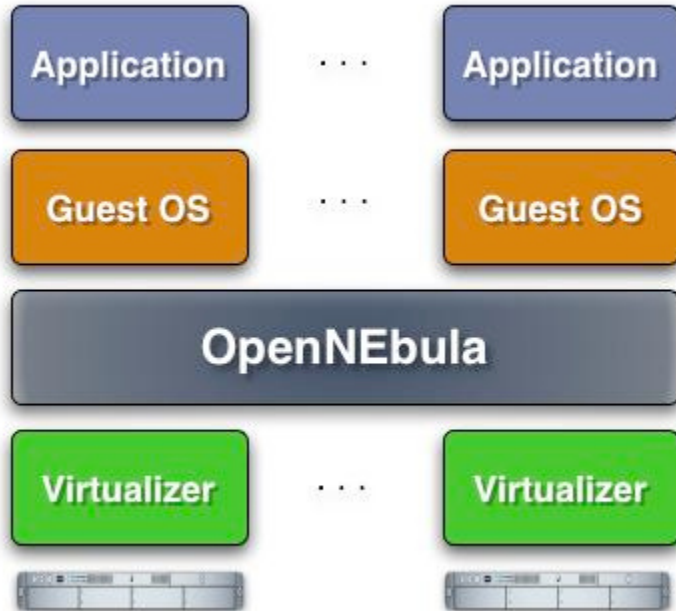
Type-II

- It requires the support of an operating system
- Also called *hosted*
- Programs managed by the OS.
- Also called hosted virtual machine.
Example: KVM, Virtual Box

Distributed Management of VMs

Extending the Benefits of Virtualization to a Physical Cluster

- VM Managers creates a distributed virtualization layer
 - Extend the benefits of VM Monitors from one to multiple resources
 - Decouple the VM (service) from the physical location



Benefits of VM Managers

- Centralized management
- Balance of workload
- Resource Optimization
- Dynamic resizing of the infrastructure
- Dynamic cluster partitioning
- Support for heterogeneous workloads
- On-demand provision of VMs
- Isolation Security
- Scalability
- Fault Tolerance
- High availability
- Cost Optimization

Limitations of VMs (Type –I and II)....

- Cold start

- Need more storage

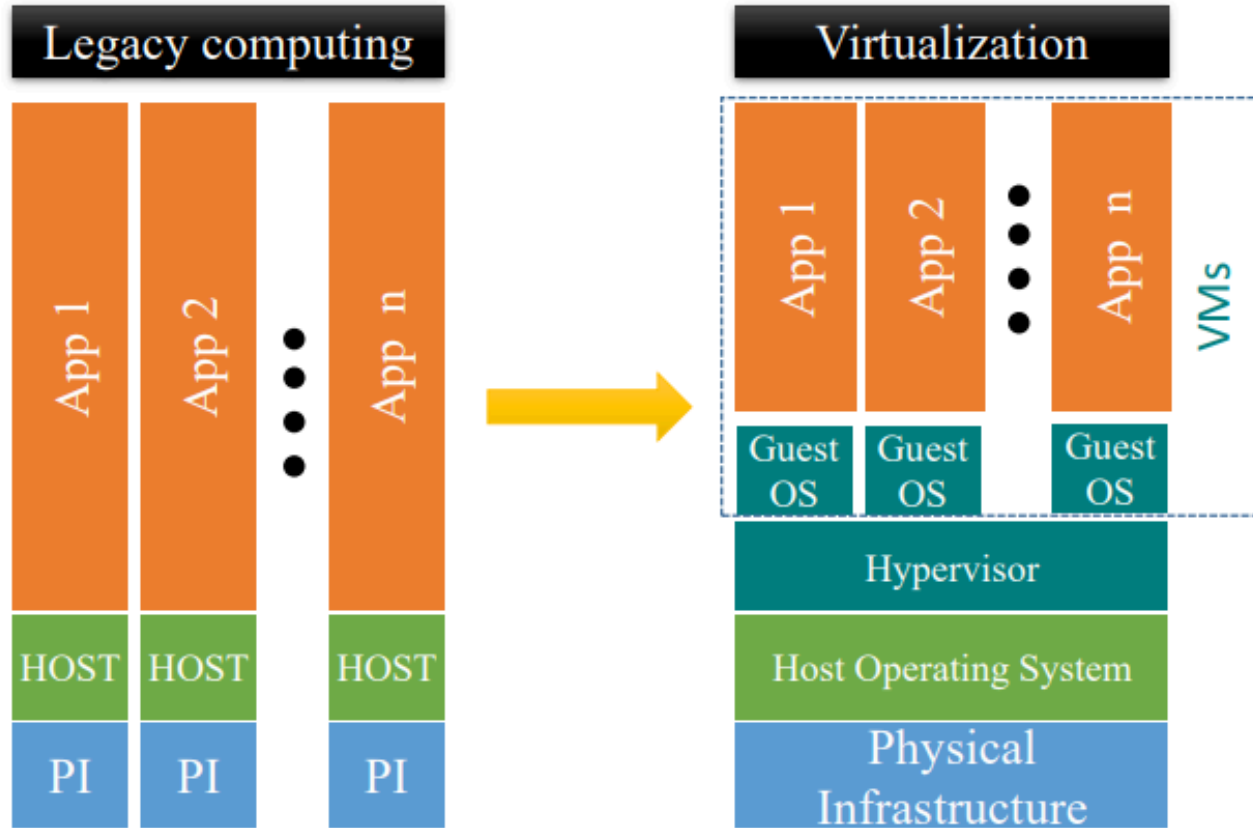
 - Less number of virtual machines per physical machine

- Performance degradation

- Inefficiency and degraded user experience

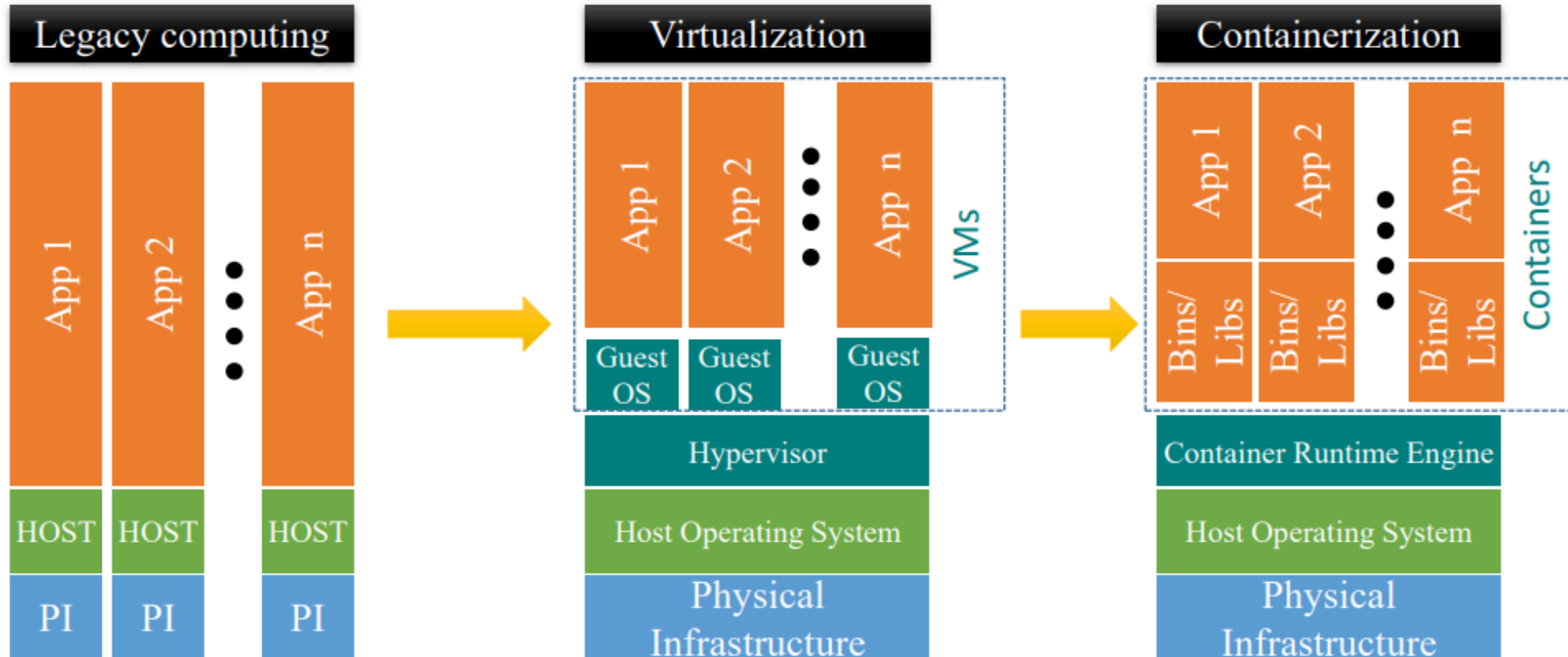
- As it interposes and abstraction layer between guest & host.

Journey of Virtualization



- But VMs are suffering from
- Cold start
 - Need more storage
 - Less number of virtual machines per PI

Containerization...



- Lightweight
- Require less memory space
- Fast lunch time
- better resource utilization
- 1000s of containers can be loaded onto a Host

- Containerized apps share Host OS's kernel to execute work

Containers